Mini-project 3 in Advanced Machine Learning (02460)

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Model

We model MUTAG adjacency matrices (Debnath et al., 1991) with a node-latent variational graph autoencoder (Kingma and Welling, 2014; Kipf and Welling, 2016). The encoder maps the seven atom labels to hidden states and applies four rounds of sum-aggregation message passing:

$$h_i^{r+1} = h_i^r + u_r\left(\sum_{j\in\mathcal{N}(i)} m_r(h_j^r)\right),$$

where m_r and u_r are ReLU MLPs. Linear heads give $(\mu_i, \log\sigma_i^2)$ and $q_\phi(Z \mid X, A) = \prod_i \mathcal{N}(z_i; \mu_i, \operatorname{diag}\sigma_i^2)$. The decoder is an inner-product link model,

$$p_\theta(A_{ij} = 1 \mid Z) = \sigma(z_i^\top z_j + b), \qquad i < j,$$

and samples graphs by drawing N from the empirical training-size distribution, $z_i \sim \mathcal{N}(0, I)$, Bernoulli upper-triangle edges, and symmetrizing. Training minimizes

$$\mathcal{L} = \mathrm{BCE}_w(A, \hat{A}) + \beta\, D_{\mathrm{KL}}(q_\phi(Z \mid X, A)\|p(Z)),$$

excluding padded nodes and self-loops. We used $d_h = 64$, $d_z = 16$, $\beta \le 0.1$ with 100-epoch warmup, Adam at 10^{-3}, learning-rate decay 0.995, 500 epochs, and positive-edge reweighting for sparsity; at sampling we subtract $\log(w_+)$ from logits to undo the density shift from this weighting.

Baseline and evaluation

The baseline is the required Erdős–Rényi model (Erdős and Rényi, 1959): sample N empirically, estimate r_N from all training graphs with that N, then draw $G(N, r_N)$. For each model we generated 1000 graphs. Novelty and uniqueness use NetworkX Weisfeiler–Lehman hashes (Shervashidze et al., 2011) against the 100 training graphs. These hashes test memorization and collapse; the graph-statistic histograms carry most of the sample-quality signal.

Table 1: WL novelty/uniqueness over 1000 generated graphs.

Method	Novel	Unique	Novel+unique
Baseline	100.0%	99.9%	99.9%
VGAE	100.0%	100.0%	100.0%

Results

Both models avoid exact memorization under the WL hash, so the more meaningful comparison is structural. We density-calibrate the VGAE prior after correcting for weighted BCE; this keeps the expected edge count comparable to the baseline while preserving the decoder's learned edge ranking. Fig. 2 shows that the VGAE matches the degree scale without relying on independent edges, but it still over-produces triangles and disconnected pieces. The result is therefore not that the VGAE uniformly dominates ER, but that the histogram and sample plots expose what the simple inner-product decoder learns and where it fails.

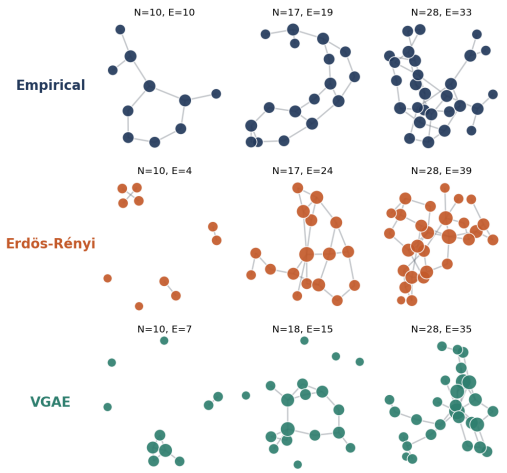

Figure 1: Representative samples; node size is degree

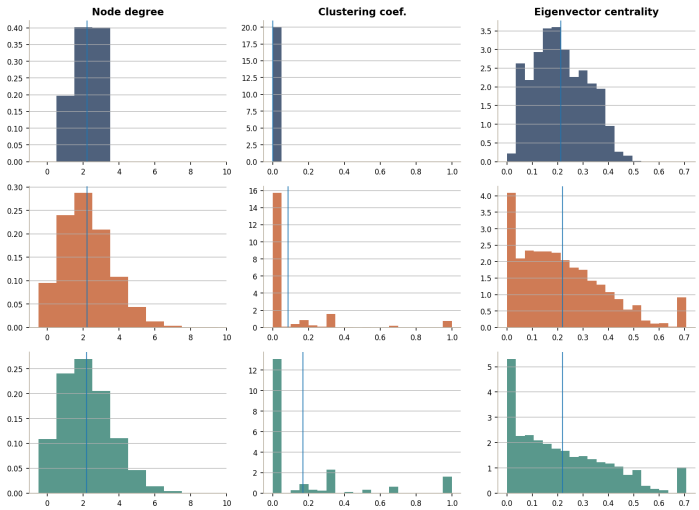

Figure 2: Histograms over 1000 samples (rows top to bottom: empirical, ER, VGAE-inner, VGAE-MLP), shared bins per column

Code snippets

Declaration of use of generative AI

- I/we have used generative AI tools: yes

What did you use the tool(s) for?

At what stage(s) of the process did you use the tool(s)?

How did you use or incorporate the generated output?

References

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